

Consonant Clusters in Georgian

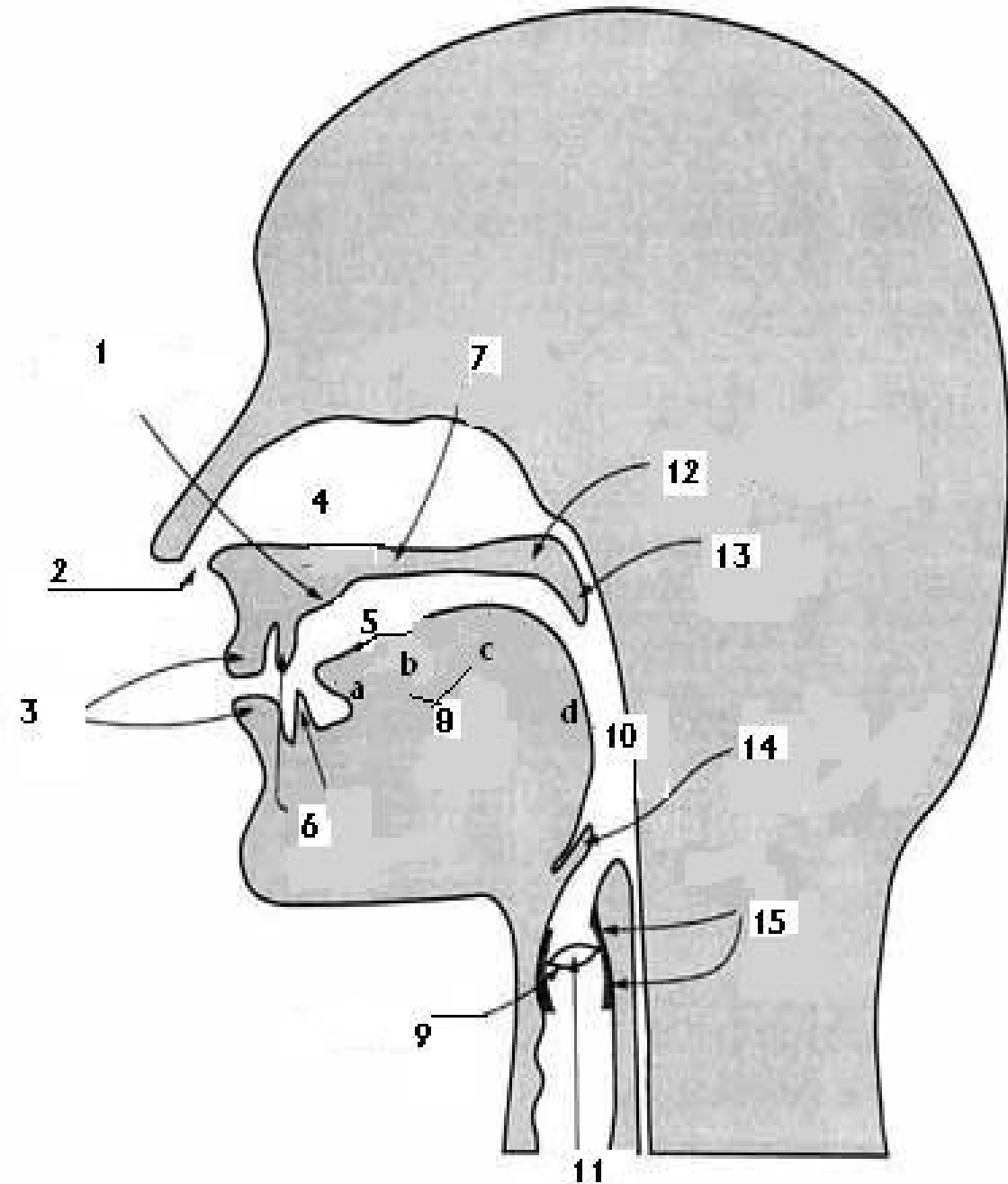
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Sagittal Section of the Vocal Tract



Classification of the Georgian consonants

Place	voicing	Voiced			Aspirated		Ejective
		plosive/ affricate	fricative	sonorants	plosive/ affricate	fricative	plosive/ affricate
Bilabial		ბ /b/		მ /m/	პ /p ^h /		პ' /p'/
labiodental			ვ /v/				
Dental		დ /d/		ნ /n/	თ /t ^h /		თ' /t'/
Alveolar		ძ /dz/	ზ /z/	ლ /l/ რ /r/	ც /ts ^h /	ს /s/	ც' /ts'/
Postalveolar		ჯ /dʒ/	წ /ʒ/		ჩ /tʃ ^h /	შ /ʃ/	ჩ' /tʃ'/
Velar		გ /g/			ქ /k ^h /		კ' /k'/
Uvular			ღ /χ // ɸ/			ხ /x // χ/	ყ /q'/
Glottal						ჲ /h/	

Overview of Georgian Consonant Clusters

- The Georgian consonantal system operates on a three-way laryngeal opposition for stops and affricates: voiced, voiceless aspirated, and voiceless ejective;
- The phonotactics permit the formation of extensive and complex consonant clusters across all word positions (initial, medial, and final);
- Initial consonant groups range from 2 to 6, generating a combinatorial inventory of over 700 permissible initial sequences in the language;
- Despite this combinatorial complexity, the overall statistical frequency of consonants versus vowels in natural text is surprisingly low, strictly constrained by high-frequency CV syllables and grammatical vowel markers;

(Vogt 1958: 6-7, 9-10; Vogt 1971: 14-15; Chitoran 1998: 122; Boeder 2005: 7-10).

The Typology of “Harmonic Clusters”

- Harmonic clusters are defined as ordered two-member obstruent pairs (C_1, C_2) where the laryngeal variable (voice, aspirate, or ejective) is strictly homogeneous;
- Distributionally, these homogeneous ordered pairs behave almost identically to simple, singular phonemes;
- Articulatorily, they are constrained by a strict regressive spatial sequence: an anterior constriction (labial or coronal) is necessarily followed by a posterior constriction (velar or uvular);
- The main subtypes include corono-velar (e.g., dg , t^hk^h (თქ)), corono-uvular (e.g., dk (დკ), $t'q'$ (ტყ)), labio-velar (e.g., bg (ბგ)), and labio-uvular (e.g., $p'q'$ (პყ));

a. Voiced

bgera ბგერა ‘sound’
bkavili ბკავილი ‘to shout’
midzɣvna მიძღვნა ‘dedication’
sadguri სადგური ‘station’
dke დკე ‘day’

b. Aspirated

p^hk^h *vili* ფქვილი ‘flour’
 sit^h *xe* სითხე ‘fluidity’
 p^h *xizeli* ფხიზელი ‘sober, awake’
 t^h *ɣovna* თხოვნა ‘request’
 ts^hk^h *era* ცქერა ‘looking’

c. Ejective

$p'k'$ *ureba* პკურება ‘to sprinkle’
 $sit'q'$ *va* სიტყვა ‘word’
 $ts'q'$ *ali* წყალი ‘water’
 $gadats'q'$ *vet'a* გადაწყვეტა ‘solution’
 $t'k'$ *ivili* ტკივილი ‘pain’

(Vogt 1958: 30-33; Vogt 1971: 14; Chitoran 1998: 123)

Phonological & Articulatory Characteristics

- Beyond laryngeal homogeneity and regressive spatial ordering, true harmonic clusters are defined by a strict formal constraint prohibiting optional [r]-insertion between C_1 and C_2 ;
- Synchronically, these clusters exhibit incredibly tight phonological cohesion;
- Distributionally, these tightly bound harmonic sequences function and behave almost identically to simple, singular phonemes within the language's morphosyntactic structure.

(Zhghenti 1956: 67, 93; Chitoran 1998: 124-125)

Structural Classification of Consonant Clusters

- Hans Vogt classifies Georgian biconsonantal clusters based on the spatial direction of their articulation;
- **Introverted Clusters (Introvertis):** The point of articulation shifts progressively backward, moving opposite to the expelled airflow (e.g., *Labial* → *Dental*, *Labial* → *Posterior*, *Dental* → *Posterior*);
 - Examples: *dg* (ღგ), *t^hk* (თკ), *dk* (დკ);
- **Extroverted Clusters (Extrovertis):** The point of articulation moves progressively forward (e.g., *Posterior* → *Dental*, *Posterior* → *Labial*, *Dental* → *Labial*);
 - Examples: *gd* (გდ), *χd* (ხდ), *t'b* (ტბ);
- Vogt further cross-categorizes these spatial vectors by their laryngeal mode, defining them as Homogeneous (sharing the exact same mode: voiced, aspirated, or glottalized) or Non-Homogeneous.

(Vogt 1958: 30-33)

Syntagmatic Constraints on Clusters

- Tedo Uturgaidze defines strict syntagmatic limits for consonant combinations operating within the boundaries of a single morpheme;
- **Prohibition of Identical Consonants:** Georgian phonotactics completely block complexes of identical consonants within a single morpheme in word-initial position;
- **Class-Based Homorganic Restrictions:** Consonants are divided into local classes based on their active articulatory organs:
 - **Class 1 (*Labials*):** /b, p', p^h, m, v/;
 - **Class 2 (*Coronals*):** /d, t', t^h, dz, ts', ts^h, z, s, dz, t', tʃ^h, ʒ, ʃ/
- A fundamental phonematic rule dictates that members of the exact same class cannot combine with each other word-initially within the same morpheme;
- Examples: Sequences like *m-b*, *p^h-m*, or *b-p^h* (Labial + Labial) are structurally prohibited as initial clusters within isolated morphemes.

(Uturgaidze 1976: 76-78)

Acoustic Evidence vs. Phonological Representation

- Modern acoustic studies contradict the traditional claim of simultaneous closures; waveforms and spectrograms clearly show two distinct release bursts for harmonic clusters;
- For instance, in words like *dgoma* *ḡḡmḡḡ* ‘standing’ or *t^hk^hma* *oŋḡḡ* ‘to say’, both the initial and medial stops display a separate, measurable closure and release;
- Furthermore, durational measurements reveal that word-medial harmonic clusters are just as long as – and in the voiced stop series, significantly longer than – standard consonant sequences formed across word boundaries;
- Consequently, acoustic data does not support treating these clusters as single complex segments; rather, they are phonetically realised as sequences of independently articulated segments;

(Chitoran 1998: 131-132, 135, 138, 140).

Extreme Consonant Clusters: The Maximal Edge

- While basic initial clusters range from 2 to 6 consonants, Georgian's agglutinative morphology permits extreme combinatorial sequences strictly at morpheme boundaries, overriding the phonotactic constraints that apply within a single morpheme;
- The maximal number of consonants at the edge of morphemes in a compound word is 11:
 - **The Example:** *vep^hχvt^hmbrdɸvneli ვეფხვთმბრდფნელი*;
 - **Meaning:** An artificially created, yet fully grammatical word meaning '(the person) who plucks the tiger/panther';
- **Theoretical Implications:** Such extreme forms raise critical questions regarding syllabification, the acoustic reality of morpheme boundaries, and the complex interaction between phonological constraints and morphological organisation.

(Vogt 1958: 6; Uturgaidze 1976: 76-78)

Exercise 1: Harmonic vs. Non-Homogeneous Identification

- **Task:** Classify the initial clusters as either harmonic (homogeneous) or non-homogeneous, justifying their answers based on laryngeal features and place of articulation.
- **Data set:**
 - *t'k'ivili* ტკივილი 'pain'
 - *dgoma* დგომა 'to stand'
 - *t'k'bili* ტკბილი 'sweet'
 - *χvd-* ხვდ- 'to meet'
 - *k'bili* კბილი 'tooth'
 - *t'ba* ტბა 'lake'

Exercise 2: The [r]-Insertion Diagnostic

- **Task:** Apply the [r]-insertion test to distinguish true harmonic clusters from standard consonant sequences.
- **Data set:**
 - *gdzeli* გძელი ‘long’
 - *bgera* ‘sound’

Exercise 3: Spatial Trajectories in Complex Clusters

Task: Analyze the initial consonant sequences in the provided lexical items. Using Vogt's structural classification; map the articulatory trajectory for each adjacent consonant pair to determine if the directional shift is **Introverted** (moving backward) or **Extroverted** (moving forward).

Data set:

- *t^hbili* თბილი 'warm'
- *χt'oma* ხტომა 'to jump'
- *dgoma* დგომა 'to stand'
- *t^hk^hven* თქვენ 'you [plural]'

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Polypersonal Agreement & Argument Structure in Georgian

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Grammatical Relations vs Semantic Roles

- **Establishing the Analytical Framework**
- To understand Georgian agreement, we must separate syntax from semantics:
 - **Grammatical Relations** (Syntactic): Subject (SUBJ), Direct Object (DO), Indirect Object (IO);
 - **Semantic Roles** (Thematic): Agent (initiator), Patient/Theme (affected entity), Experiencer (perceiver), etc.
- **The Polypersonal/Bipersonal Nature of Georgian:**
 - Georgian is typically classified as a polypersonal language;
 - However, due to strict morphological slot constraints, the verb practically functions with a bipersonal restriction, surfacing a maximum of two overt person markers at a time;
 - **Why “Subject” and “Object” fail:** A semantic Experiencer might be the syntactic Subject, yet be indexed by morphological “Object” markers. For instance, in “verbes invertis” (inverted verbs), the singular/plural opposition of the formal subject is neutralized, and the indirect object is interpreted as the logical subject; **OR THE OPPOSITE VIEW OF SUBJECTHOOD IS ALSO CONSIDERED;**

(Harris 1981: 117-118, 127-128; Vogt 1971: 135-136)

The Marker Sets

The v-set and the m-set

Georgian agreement relies on two competing paradigms. Note: Anderson (1984) models these as morphological feature-blocks, whereas Harris (1981) links them to initial/final syntactic strata.

Person	v-set (Canonical Subject)	m-set (Canonical Object)
1 st SG	<i>v-</i>	<i>m-</i>
1 st PL	<i>v- ... -t</i>	<i>gv-</i>
2 nd SG	<i>∅ / x-</i>	<i>g-</i>
2 nd PL	<i>∅ / x- ... -t</i>	<i>g- ... -t</i>
3 rd SG	<i>-s, -a, -o</i>	<i>-s, -h, -∅</i>
3 rd PL	<i>-en, -an, -es, -nen</i>	<i>-s, -h, -∅ ... -t</i>

General Distribution:

- **v-set:** Typically indexes Subj in Series I (Present) and II (Aorist).
- **m-set:** Typically indexes DO and IO.

(Anderson 1984: 161; Vogt 1971: 80-81)

Georgian

1. **Bipersonal agreement** – subject & direct object; subject & indirect object – *monotransitive* ~ *ditransitive*;
2. **Split Ergativity** – defined by TAM (NB: **Screeve** appears in the Grammatical Features Inventory (<https://www.smg.surrey.ac.uk/features/morphosemantic/screeve/>));
- 3 . **Patterns (bivalent transitive verbs):**
 - Present** – NOM DAT (arguments) + personal/number markers (verb)
 - Aorist** – ERG NOM (arguments) + personal/number markers (verb)
- 4 . **Patterns (trivalent transitive verbs):**
 - Present** – NOM DAT DAT (arguments) + personal/number markers (verb)
 - Aorist** – ERG NOM DAT (arguments) + personal/number markers (verb)

Subject agreement markers

	Singular	Plural
1 st	<i>v-</i>	<i>v-...-t</i>
2 nd	<i>x-, Ø-</i>	<i>x-...-t // Ø-...-t</i>
3 rd	<i>-s // -a // -o</i>	<i>-n // -en // -an // -nen // -es</i>

Object agreement markers

	Singular	Plural
1 st	<i>m-</i>	<i>gv-</i>
2 nd	<i>g-</i>	<i>g-...-t</i>
3 rd	<i>h- // s- // Ø-</i>	<i>h-...-t // s-...-t // Ø-...-t</i>

(Shanidze 1980; Kvachadze 2019: 16-19, 19 - 25; Aronson 1990: 40-41, 110-111; Hewitt 1995: 3, 117-125, 128)

Combinability and compatibility of plural marking

Subject plurality “oppresses” the object plurality.

თქვენ	ცნობილ-ი	მხატვარ-ი	გ-ხატ-ავ-თ
<i>tkven</i>	<i>tsnobil-i</i>	<i>mxat'var-i</i>	<i>g-xat'av-t</i>
2PL	famous-NOM	painter-NOM	2OBJ-paint-THEM-PL
“A famous painter is painting you.”			

თქვენ	მხატვრ-ებ-ი	გ-ხატ-ავ-ენ
<i>tkven</i>	<i>mxat'vr-eb-i</i>	<i>g-xat'av-en</i>
2PL	painter-PL-NOM	2OBJ-paint-THEM-3PL.SBJ
“Famous painters are painting you.” (Arabuli 2008: 165-166).		

Quirky Case: Subject Marker Dropping

<i>(მე)</i>	<i>(შენ)</i>	<i>ძმ-გ-კლ-ავ</i>
<i>(me)</i>	<i>(fen)</i>	<i>mo-g-k'l-av</i>
1SG	2SG	PREV-2OBJ-kill-THEM
“I will kill you”. [spelling and glosses are modified]		

“It is argued in Anderson 1984b and 1986a that the non-appearance of /v-/ is a morphological fact, to be accounted for by a disjunctive relationship among the possible prefixes in the structure of the Georgian Verb”.

the morphological rules of inflection are organized into **blocks**, where the relation among rules within the same block is a **disjunctive** one.

(Anderson 1992: 87, 1 2 8 - 1 2 9).

Agreement in Action

Morphological Segmentation and Glossing

- **Example 1:**

- *ɔbɔdɔɔɔ vχedav*
- *v-χed-av*
- 1SBJ-see-TM
- Word-for-word: I-see-him/her/it
- Translation: 'I see him/her/it.' (The 3rd person object is unmarked/null).

- **Example 2:**

- *ɔbɔdɔɔɔ mχedav*
- *m-χed-av*
- 1OBJ-see-TM
- Word-for-word: me-see-you(SG)
- Translation: 'You see me.' (The 2nd person subject is marked by a null prefix).

(Hewitt 1995: 128-132; Vogt 1971: 85)

Slot Competition & Person Hierarchy

The Prefixal Bottleneck

- **The Constraint:** The Georgian verb possesses only one prefixal slot for person marking;
- **The Hierarchy:** Speech Act Participants (1st/2nd Person) > 3rd Person;
- **The Conflict:** What happens if a bivalent verb requires indexing both a 1st-person participant and a 2nd-person participant simultaneously?
- **The Resolution:** The two markers compete for the single prefix slot. The expected sequence of *v-g-* (for meanings like 'I kill you') is never realized. The subject-affix is dropped before the object-marker *-g-*, leaving only the object marker to surface.

(Anderson 1984: 162-163; Hewitt 1995: 131)

The Inversion Phenomenon

Class IV Verbs (Experiencer/Affective)

- Inversion proves that Georgian agreement is not a simple mapping of Nominative=Subject and Dative=Object;
- In Class IV verbs (verbs of emotion, perception, state), the argument mapping is “inverted”
- **Logical Subject** (Experiencer): Marked with Dative case, but indexed by the m-set (Object markers);
- **Logical Object** (Stimulus): Marked with Nominative case, but indexed by the v-set (Subject markers);
- **Theoretical debate**: Is this a syntactic demotion of the subject, or a purely morphological case-spell-out rule?

(Harris 1981: 117-120, 127-128; Anderson 1984: 172-177; Vogt 1971: 135-136)

Visualising Inversion

- **Crossed Argument Mapping**
 - Example: *ḏnyʒaṁb miq'vars* 'I love him/her/it';
 - Syntax & Semantics:
 - Experiencer (I): Syntactic Subject.
 - Stimulus (him/her/it): Syntactic Object.
- **Morphology** (Inverted Mapping):
 - *m-* (1st Person m-set prefix)  Indexes the logical Subject ('I');
 - *-s* (3rd Person v-set suffix)  Indexes the logical Object ('him/her/it');
 - *m-i-q'var-s* 1OBJ-PRV-love-3SBJ 'I love him/her/it';

(Hewitt 1995: 122; Harris 1981: 127; Anderson 1984: 176)

Resolving the Competition

- **Paradigm:** 1st vs. 2nd Person
 - Let us look at the verb *χedav* 'see' when both arguments are Speech Act Participants;
 - Combinations such as "I-you", "you-me", "we-you", and "you-us" are systematically excluded from surfacing with two explicit prefixes;
 - 'You (SG) see me.'
- **Arguments:** 2SG Subj (Ø-), 1SG Obj (*m*-).
- **Result:** *ḡbɛɖʌɟ* (*m-χedav*). The 1st person object prefix surfaces;
 - 'I see you (SG)'
- **Arguments:** 1SG Subj (*v*-), 2SG Obj (*g*-).
 - Competition: *v*- vs. *g*-
- **Result:** *ɟbɛɖʌɟ* (*g-χedav*). The 2nd person object marker (*g*-) wins the prefix slot. The 1st person subject marker (*v*-) is deleted/suppressed.

(Anderson 1984: 162; Vogt 1971: 84-85)

Exercise 1: Interactive Data-Analysis Exercise

Determine Controllers and Grammatical Relations

- Analyse the following three forms. For each, identify:
- The morphological segmentation (prefix vs suffix);
- The agreement controllers (which argument controls which affix?);
- The winning prefix (if there was slot competition);
- **Form A:** *გხატავს gḡat'avṣ* 'He/she paints you (SG)'
- **Form B:** *გხედავ gḡedav* 'I see you (SG)'
- **Form C:** *ვმალავთ ṽmalavt* 'We hide him/her/it'

Exercise 2: Slot Competition & Prefix Suppression

Task: Examine the interaction of 1st-person subject and 2nd-person object markers in Georgian to determine how the language handles morphological collisions;

Data set:

Form A: *v-χat'av* 'I paint him/her/it' (1SBJ-paint)

Form B: *g-χat'av-s* 'He/she paints you' (2OBJ-paint-3SBJ)

Form C (Ungrammatical): **v-g-χat'av* or **g-v-χat'av* (Intended: 'I paint you')

Form D (Actual): *g-χat'av* 'I paint you'

Questions for Students:

1. Based on Forms A and B, what are the expected prefixes for a 1st-person Subject and a 2nd-person Object?
2. Why is the hypothetical Form C strictly ungrammatical in Georgian, and what rule resolves this collision to produce Form D?

Answer

1. Expected Prefixes: The 1st-person subject requires the *v*- prefix (from the *v*-set), and the 2nd-person object requires the *g*- prefix (from the *m*-set);
2. The Prefixal Bottleneck: Georgian verb morphology strictly limits the verb to only one prefixal slot for person marking. Therefore, a sequence like **v-g*- is structurally blocked from surfacing;
3. The Resolution Rule: When the 1st-person subject marker (*v*-) and the 2nd-person object marker (*g*-) compete for the single prefix slot, the object marker (*g*-) wins. The subject marker (*v*-) is systematically suppressed or deleted before the object marker;
4. **Conclusion:** Thus, the morphologically correct form *g-χat'av* translates to 'I paint you', with the 1st-person subject implied by the suppression of the 3rd-person suffix.

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Morphosyntactic interaction

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Morphosyntactic Interaction and Verb Classes

The Intersection of Screeves and Lexical Categories

- The Georgian verbal system features a profound interaction between tense/aspect/mood (organised into three “screeve” series), case marking, and polypersonal agreement;
- Morphosyntactic alternations are intrinsically linked to the traditional classification of Georgian verbs into four classes, which govern their behaviour across paradigms;
- **Class I** encompasses transitive verbs;
- **Class II** includes intransitive and passive verbs;
- **Class III** comprises active intransitive (medioactive) verbs, which often denote atelic activities, and **Class IV** consists of indirect or experiencer verbs;
- **Examples of Classes:** Class I *da-c'er-s* ‘will write’; Class II *da-i-c'er-eb-a* ‘will be written’; Class III *i-tsekv-eb-s* ‘will dance’; Class IV *u-q'var-s* ‘loves’;
- Modern morphological frameworks sometimes regroup these into broader categories, distinguishing “active” stems (Classes I and III) from “passive” or “mediopassive” stems (Classes II and IV) based on shared case-assigning and inversion properties;

(Harris 1981: 2; Anderson 1984: 159-160, 166, 174; Hewitt 1995: 217-392; Tuite 1998: 15; Tuite 2017: 1123)

Alignment Patterns in Series I and II

From Nominative to Split-Ergative Systems

- Georgian displays distinct alignment patterns across its three screeve series, fundamentally altering the morphological case of core arguments;
- In Series I (Present/Future subseries), the language operates on a nominative-accusative pattern, where the agent of a transitive verb takes the nominative case and the patient takes the dative case;
- **Example** (Series I): *K'ats-i iḡat'-av-s surat-s* 'The man [NOM] paints a picture [DAT]';
- In Series II (Aorist subseries), the alignment shifts to a split-ergative, or active, pattern;
- Consequently, in Series II, the agent of a transitive verb (Class I) and the subject of an active intransitive verb (Class III) are marked with the ergative case;
- The patient of the transitive verb in Series II shifts to the nominative case, aligning its case marking with the subjects of inactive intransitive verbs in Class II;
- **Example** (Series II): *K'ats-ma da-ḡat'-a surat-i* 'The man [ERG] painted a picture [NOM]'.

(Harris 1981: 1-2; Anderson 1984: 164, 166-167; Boeder 2005: 60)

Series III and the Phenomenon of Inversion

The Realignment of Experiencers and Agents

- In Series III (Perfect series), transitive and active intransitive verbs (Classes I and III) undergo a radical morphosyntactic realignment known as inversion;
- During inversion, the logical agent of the action appears in the dative case and controls object agreement markers on the verb;
- Simultaneously, the logical patient appears in the nominative case and controls subject agreement markers;
- **Example** (Series III): *k'ats-s da-u-xat'-av-s surat-i* 'The man [DAT] has painted a picture [NOM]';
- This specific inverted **mapping of arguments** to morphology is not restricted solely to Series III; it permanently characterises Class IV experiencer and affective predicates across all tense series;
- The indirect object of the original active construction loses its term status during inversion and surfaces as a postpositional phrase, typically marked with *-tvis* 'for'.

(Harris 1981: 117- 118; Anderson 1984: 172-174 ; Boeder 2005: 61)

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(Harris 1981: 117-120, 127-128; Anderson 1984: 172-177; Vogt 1971: 135-136)

Visualising Inversion

- **Crossed Argument Mapping**
 - Example: *ፈንገሳጽ ማባላላ* *miq'vars* 'I love him/her/it';
 - Syntax & Semantics:
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(Hewitt 1995: 122; Harris 1981: 127; Anderson 1984: 176)

Subject and object marking

<i>k'ats-i</i>	<i>da-χat-av-s</i>	<i>m egobar-s</i>
man-NOM	PV-draw-TM- 3 SG.SBJ	friend-DAT
'The man will draw his friend.'		

<i>k'ats-ma</i>	<i>da-χat-a</i>	<i>m egobar-i</i>
man-ERG	PV-draw-3 SG.SBJ.AOR	friend-NOM
'The man drew his friend.'		

<i>k'ats-s</i>	<i>da-u-χat-av-s</i>	<i>m egobar-i</i>
man-DAT	PV-PRV-draw-TM- 3 SG.SBJ.AOR	friend-NOM
'The man has drawn the friend.'		

da-u-χat-av-s – grammatical homophony!

Theoretical Implications of Split Ergativity and inversion

Morphological Spell-out vs. Syntactic Movement

- The dramatic alignment shifts across the screeve series raise a core theoretical question: is Georgian split ergativity primarily a matter of surface morphological case marking, or does it carry deep syntactic consequences?
- Within the framework of Relational Grammar, inversion is analysed as a syntactic rule that demotes the initial subject to an indirect object, while an unaccusative rule promotes the initial direct object to a final subject;
- Conversely, alternative morphological frameworks argue that inversion does not alter the underlying syntactic structure of the clause;
- Under the A-Morphous morphological view, inversion is strictly a rule operating on layered morphosyntactic representations within the inflectional component, manipulating dummy agreement layers without syntactic movement;
- The determination of grammatical subjecthood remains a complex debate, as different Georgian dialects show varying degrees of morphosyntactic orientation toward semantic subjects versus absolute/nominative arguments.

(Harris 1981: 118; Anderson 1984: 180-182, 190-192; Anderson 1992: 137-156; Tuite 1998: 212-214; Tuite 2017, 1132-1135)

Task 1: Alignment Tracking across Screeves

Let's take the Class I verb root *-ts'er-* 'write' and the arguments *k'ats-* 'man' and *ts'eril-* 'letter';

- Construct three simple clauses using these elements in the Present (Series I), Aorist (Series II), and Perfect (Series III) screeves;
- **Analysis:** Identify the morphological case of both arguments in each series and explicitly state which alignment pattern (Nominative-Dative, Ergative-Nominative, or Dative-Nominative) is operating in each instance.

Task 2: Find the Pattern

Examine the three sentences:

A. *k'atsi gat'exs tf'ikas*

B. *k'atsma gat'exa tf'ika*

C. *k'atss gaut'exia tf'ika*

1. Identify the specific morphological case suffixes attaching to the agent (the man) and the patient (the glass) in clauses A, B, and C.
2. Does the logical subject's // agent's semantic role change across the three clauses? Does its morphological case remain constant?
- 3 .What changes from $A \rightarrow B \rightarrow C$?

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